18 марта 2026 г. 18:54

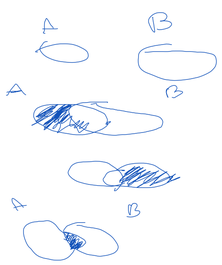
ins

inst.db.schema.table_nam

use db_name

select from sb.table

Left
right
outer



как БД физические

Key_A Key_B

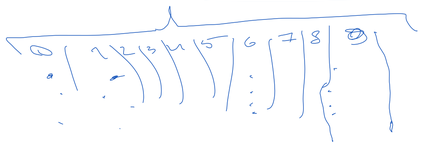
HashJoin:

$V(A) < V(B)$

Key_A

hash map

$\forall Key_i \rightarrow hash(Key_i) \,//\, 10$



$\exists Key_j = 3$

вместо $\rightarrow$ res

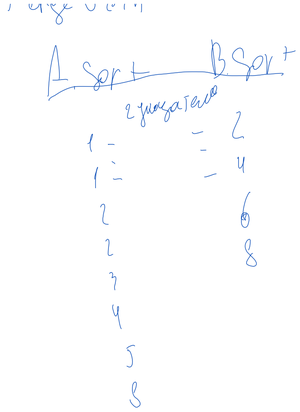
$a_{key_1} >= b_{key_2}$

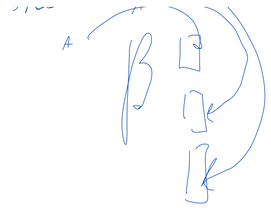
Nested loop

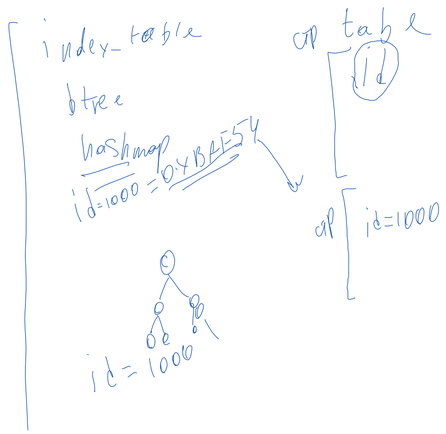
$O(n \times m)$ — долго, и дорого

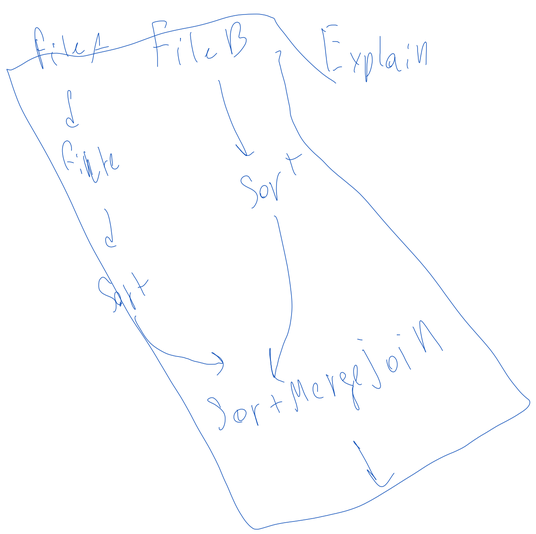
Merge Join

Broadcast

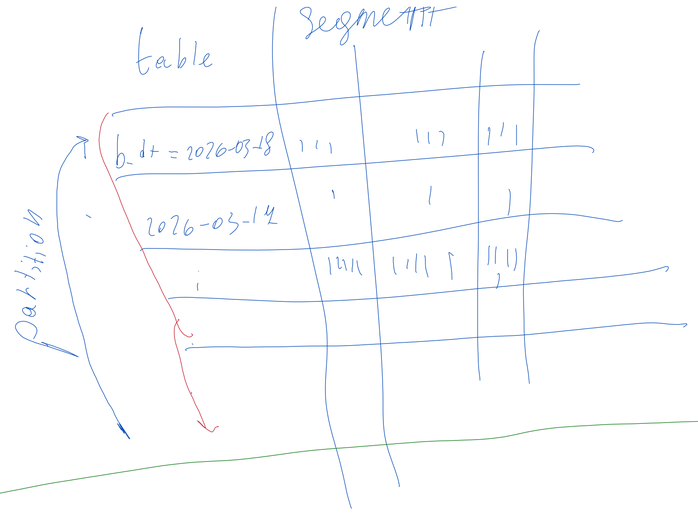




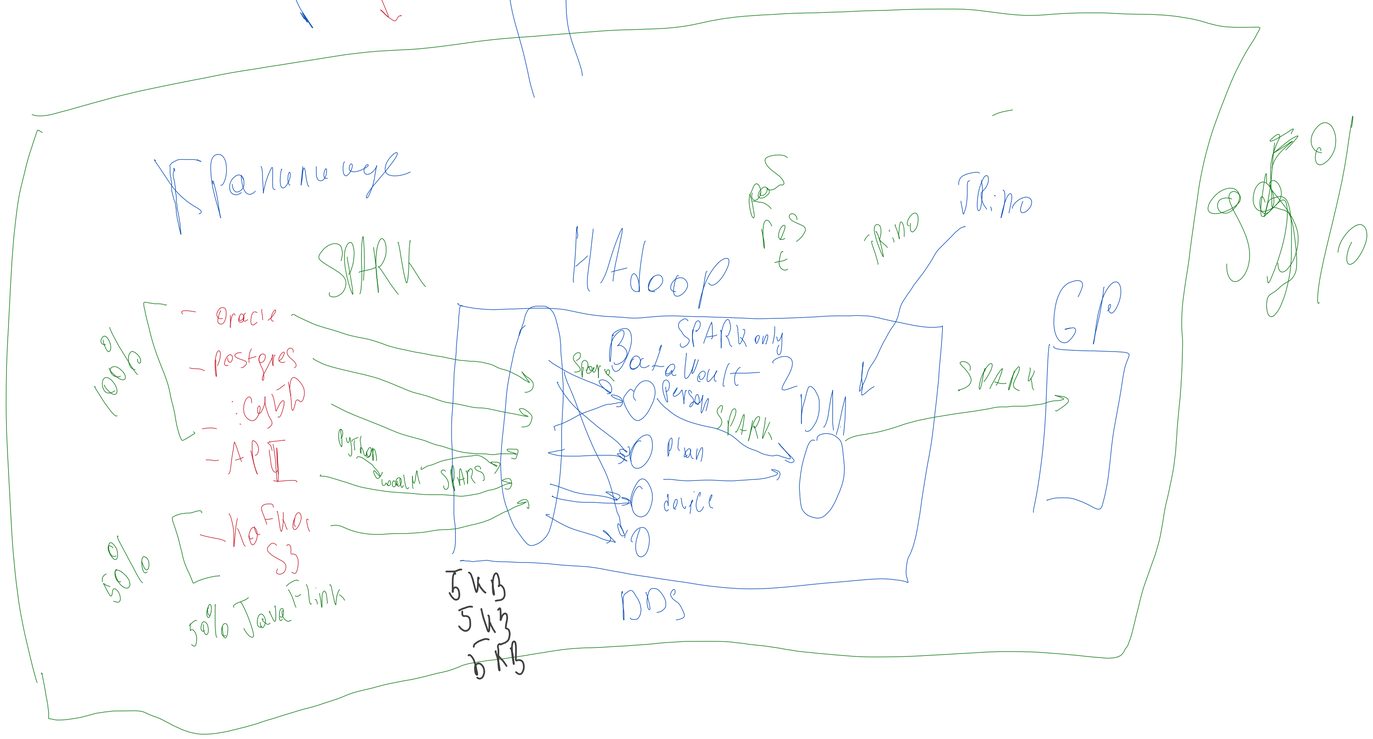




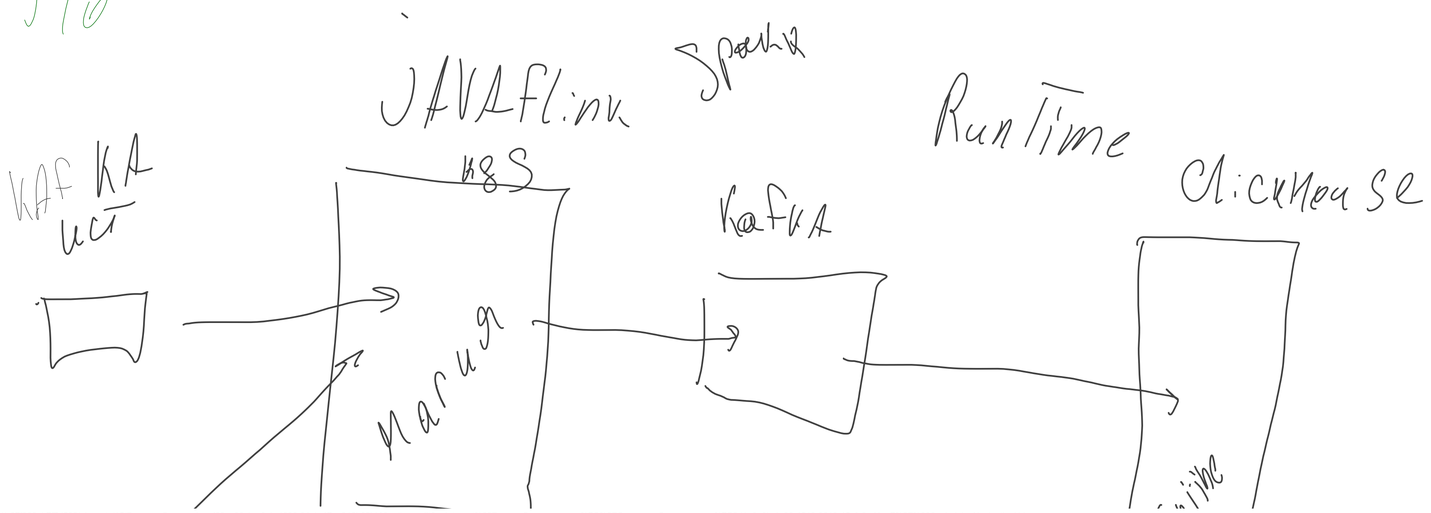
Analyze Sh. table



Create table
⋮
Partitionby(b_dt)
DistributedBy
(msisdn)



5%



Pic

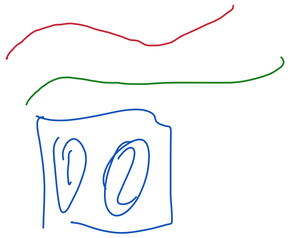
Yafke Eff

SPARK Catylst.

Руками

1. ↓ wrk
2. Repatitio
3. разбить
4. bucket

Resource



spark-nlst SPARK-Tuning

и сп
входные

LLM

RAG → выделять лучше

Plan

LLM

plan чиз 10 43 с

50

1

Docker

DBeaver

Table1 → Table2

Big Data → ML

ETRL